

NORWOOD, MA, USA

WMO: 725098

Lat: 42.191N

Lon: 71.174W

Elev: 50

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 54704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	9.0	-11.0	3.0	9.0	-6.7	3.8	13.0	24.5	30.2	22.0	34.2	4.3	300	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	91.0	73.6	88.3	72.4	85.3	70.6	76.9	86.6	75.3	83.4	73.7	81.0	9.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	125.9	80.7	72.6	121.2	79.6	71.7	117.2	78.6	40.2	86.6	38.7	83.6	37.3	80.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.1	16.1	DB	-4.9	95.4	7.2	3.0	-10.1	97.6	-14.3	99.3	-18.3	101.0	-23.5	103.1	(4)
			WB	-5.6	79.2	7.0	1.8	-10.6	80.5	-14.7	81.6	-18.7	82.6	-23.8	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	27.7	29.9	36.9	48.1	57.9	66.6	72.3	70.6	63.2	51.9	42.5	33.5	(6)
(7)		DBStd	17.46	10.76	9.73	8.96	8.19	7.78	7.00	5.97	6.08	7.45	8.23	8.45	9.43	(7)
(8)		HDD50	2667	693	567	418	130	13	0	0	1	74	254	517	(8)	
(9)		HDD65	6046	1156	984	871	510	250	63	9	18	121	414	674	976	(9)
(10)		CDD50	2745	2	3	13	74	258	499	691	638	397	133	30	7	(10)
(11)		CDD65	647	0	0	0	4	29	112	236	190	67	9	0	0	(11)
(12)		CDH74	6760	0	0	7	83	412	1167	2457	1937	626	70	1	0	(12)
(13)		CDH80	2293	0	0	1	27	133	383	931	644	166	8	0	0	(13)
(14)	Wind	WSAvg	6.4	6.9	7.2	7.9	7.6	6.5	6.0	5.6	5.0	5.0	5.9	6.2	6.4	(14)
(15)	Precipitation	PrecAvg	47.5	3.9	3.4	4.6	4.1	3.7	3.9	3.4	3.9	3.8	4.4	3.9	4.5	(15)
(16)		PrecMax	61.1	7.2	8.1	15.4	9.0	9.6	13.2	7.8	7.9	9.3	13.3	6.7	8.6	(16)
(17)		PrecMin	32.6	1.6	1.3	0.3	1.2	1.2	0.2	1.4	0.8	0.5	0.1	1.0	1.5	(17)
(18)		PrecStd	7.7	1.4	1.5	3.0	1.9	1.9	3.3	1.5	1.9	2.2	3.0	1.5	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	61.6	62.5	71.2	83.9	89.9	92.8	95.4	93.5	90.2	80.7	70.3	65.2	(19)	
(20)		MCWB	57.3	54.2	56.0	62.2	68.2	74.5	77.0	76.2	73.8	66.6	61.6	58.5	(20)	
(21)		DB	55.0	54.9	62.5	73.7	83.6	88.2	91.4	89.8	84.4	75.2	65.7	58.9	(21)	
(22)		MCWB	49.8	49.0	51.5	57.1	66.5	72.5	74.2	73.7	70.4	65.5	59.2	54.4	(22)	
(23)		DB	48.4	49.7	56.0	67.8	78.5	84.1	88.4	86.5	80.9	71.4	61.8	53.7	(23)	
(24)		MCWB	43.5	43.4	47.2	54.9	63.1	69.2	72.8	71.9	68.9	63.4	55.3	49.0	(24)	
(25)		DB	43.3	44.8	51.5	63.3	73.2	81.1	85.2	83.6	77.1	67.1	57.4	48.5	(25)	
(26)		MCWB	38.4	38.9	44.0	52.6	60.8	68.1	71.2	70.6	67.1	59.8	51.5	43.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	58.5	57.2	58.9	65.1	71.6	76.9	79.2	78.3	76.1	71.4	64.9	60.9	(27)	
(28)		MCDWB	61.7	60.7	66.4	78.0	84.6	87.3	90.5	89.7	83.4	76.2	68.4	63.4	(28)	
(29)		WB	51.2	50.1	54.1	61.6	68.9	74.7	77.3	76.5	73.5	67.8	60.5	55.2	(29)	
(30)		MCDWB	53.4	54.2	60.7	70.8	79.8	84.3	87.6	85.4	79.6	72.7	63.8	58.8	(30)	
(31)		WB	44.1	44.1	49.0	58.0	66.2	72.6	75.4	75.0	71.9	64.4	56.7	50.1	(31)	
(32)		MCDWB	48.0	48.1	54.4	64.5	74.7	80.8	83.9	82.3	78.0	69.6	60.5	53.1	(32)	
(33)		WB	39.0	39.3	44.3	54.1	63.1	70.5	73.7	73.3	69.9	61.2	52.8	44.7	(33)	
(34)		MCDWB	42.6	43.6	49.9	60.9	70.1	77.5	81.1	79.9	75.2	66.2	56.4	48.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.8	19.4	18.9	21.0	22.2	21.4	21.8	22.4	23.4	22.5	20.2	17.7	(35)	
(36)		5% DB	23.2	25.0	26.7	29.4	30.3	26.1	24.7	24.8	25.7	26.6	24.6	22.5	(36)	
(37)		MCWBR	19.7	19.2	18.0	15.9	14.9	12.6	10.7	11.4	14.0	17.4	18.2	19.8	(37)	
(38)		5% WB	22.6	22.7	24.5	24.4	25.7	22.1	20.7	20.0	19.9	22.0	21.0	21.3	(38)	
(39)		MCWBR	20.4	19.7	19.0	16.1	14.2	11.7	10.1	10.3	12.4	16.5	18.0	20.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.302	0.327	0.371	0.408	0.453	0.473	0.435	0.374	0.334	0.310	0.294	(40)	
(41)		taud	2.496	2.482	2.473	2.381	2.314	2.221	2.170	2.265	2.413	2.537	2.567	2.563	(41)	
(42)		Ebn at Noon	273	288	292	285	275	262	255	262	272	273	266	264	(42)	
(43)		Edh at Noon	23	27	31	36	40	44	46	41	33	26	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	571	846	1164	1493	1656	1795	1882	1692	1365	908	621	473	(44)	
(45)		RadStd	38	73	81	160	181	173	110	81	95	80	48	50	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	N/A	(46)
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+79	(47)

Nomenclature: See separate page